Kigali Independent University (ULK)

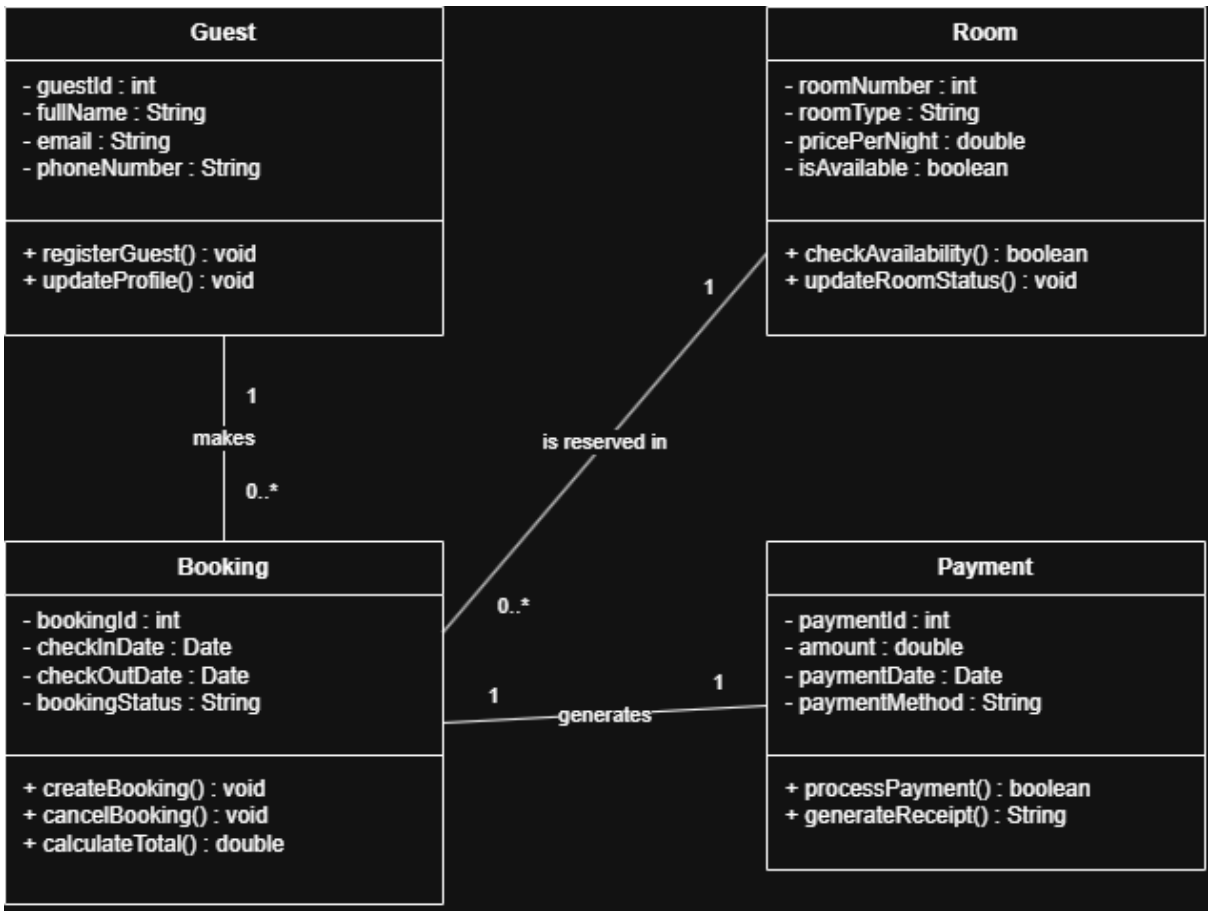
Course: UML Class Diagram Modelling Assignment
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REPORT 2: UML CLASS DIAGRAM REPORT

Project Title: Hotel Booking Management System

1. Complete UML Class Diagram:



2. Classes, Attributes, and Methods:

Class 1: Guest
- Attributes:
* - guestId : int
* - fullName : String
* - email : String
* - phoneNumber : String
- Methods:
* + registerGuest() : void
* + updateProfile() : void

Class 2: Room
- Attributes:
* - roomNumber : int
* - roomType : String
* - pricePerNight : double
* - isAvailable : boolean
- Methods:
* + checkAvailability() : boolean
* + updateRoomStatus() : void

Class 3: Booking
- Attributes:
* - bookingId : int
* - checkInDate : Date
* - checkOutDate : Date
* - bookingStatus : String
- Methods:
* + createBooking() : void
* + cancelBooking() : void
* + calculateTotal() : double

Class 4: Payment
- Attributes:
* - paymentId : int
* - amount : double
* - paymentDate : Date
* - paymentMethod : String
- Methods:
* + processPayment() : boolean
* + generateReceipt() : String

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3. Relationships and Multiplicities:
- Guest to Booking: One-to-Many (1 to 0..*) — Association.
- Room to Booking: One-to-Many (1 to 0..*) — Association.
- Booking to Payment: One-to-One (1 to 1) — Association.

4. Brief Explanation of Relationships:
- Guest — Booking (1 to 0..*): A single guest can make zero or many bookings over time, while each specific booking record belongs to exactly one guest.
- Room — Booking (1 to 0..*): A single hotel room can be reserved in zero or many bookings across different dates, and each booking is made for one specific room.
- Booking — Payment (1 to 1): Each confirmed booking generates exactly one payment record, and each payment is processed for one specific booking.